

Linear First-Order Equations & the Integrating Factor



Recognizing linear form

- 1 Write $y' + 2xy = x$ in the standard form $y' + P(x)y = Q(x)$ and identify $P(x)$ and $Q(x)$.
 - 2 Is $y' = \frac{y}{x} + x^2$ linear? Rewrite it in standard form.
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Finding the integrating factor

- 3 Find the integrating factor for $y' + 3y = e^x$.
 - 4 Find the integrating factor for $y' - \frac{2}{x}y = x$, $x > 0$.
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Solving with the integrating factor

- 5 Solve $y' + 3y = e^x$.
 - 6 Solve $y' - \frac{2}{x}y = x$, $x > 0$.
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Initial value problems

- 7 Solve $y' + y = 2$, $y(0) = 0$.
 - 8 Solve $y' + 2y = 4x$, $y(0) = 1$.
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Application: a filling tank

- 9 A tank starts with 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution flows out at the same rate. If $Q(t)$ is the salt (kg) at time t , the balance gives $Q' = 6 - 0.03Q$. Solve for $Q(t)$ given $Q(0) = 0$.
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More integrating factor practice

- 10 Find the integrating factor for $y' + \frac{1}{x}y = \cos x$, $x > 0$.
 - 11 Solve $y' + \frac{1}{x}y = \cos x$, $x > 0$.
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More initial value problems

- 12 Solve $y' - 2y = 4$, $y(0) = 3$.
- 13 Solve $y' + \frac{y}{x} = 3x^2$, $y(1) = 2$, $x > 0$.
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Application: an RL circuit

- 14 An RL circuit satisfies $L \frac{di}{dt} + Ri = V$. With $L = 2$, $R = 10$, $V = 20$, and $i(0) = 0$, solve for $i(t)$.
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A second mixing-tank variant

- 15 A 200 L tank starts with 50 kg of dissolved salt. Pure water flows in at 5 L/min and the mixture flows out at the same rate. Find $Q(t)$, the salt (kg) at time t .
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Checking a solution by substitution

- 16 Verify that $y = 3 - 2e^{-x}$ solves $y' + y = 3$.
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One more round of practice

- 17 Solve $y' + 4y = 8$, $y(0) = 5$.
- 18 Find the integrating factor for $y' + \frac{2}{x}y = 3x$, $x > 0$, and solve.
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One final integrating-factor problem

- 19 Solve $y' - 3y = 6$, $y(0) = 1$.
- 20 Find the integrating factor for $y' + 5y = e^{-2x}$.
- 21 Solve $y' + 5y = e^{-2x}$ using the integrating factor found above.
- 22 Solve $y' - y = e^{2x}$, $y(0) = 0$.
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A closing pair of mixed problems

- 23 Find the integrating factor for $y' + \frac{3}{x}y = x^2$, $x > 0$.

24 Solve $y' + \frac{3}{x}y = x^2$ using the integrating factor found above.